Network Throughput Optimization: Problem Statement

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Summary

Route the current demand on four streams through the network so that as much of it as possible reaches its exit. Two streams go to UK and two to US. In each pair, one stream is latency-sensitive and should see as little latency as possible; the other only needs to be delivered. Link bandwidth is limited, so the four streams compete for it, and the routing is solved for one instant's demand at a time.

Background

The network carries mixed traffic to two exits, UK and US. Some of it, such as a video call, is useless if it takes a slow route. The rest, such as a file transfer or buffered video, only needs to arrive.

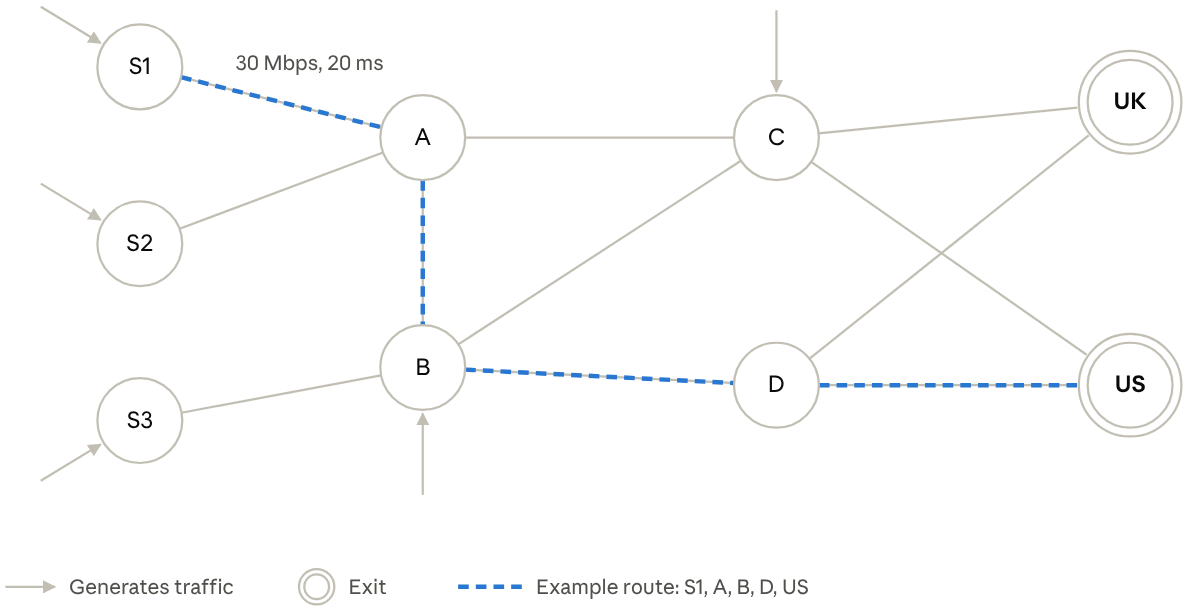
Both kinds share the same links. If tolerant traffic fills the fastest routes, latency-sensitive traffic is pushed onto slower ones. Routing each stream on its own misses this, so all four streams have to be planned together.

The network

The case to solve is a sparse network of 9 nodes with two exits, UK and US, drawn below. Five nodes generate traffic (the incoming arrows), and every exit can be reached by more than one route, so routing is a real choice.

Five nodes generate traffic, and each exit has two routes in

9-node case network; every link is full-duplex



case network · 9 nodes, 2 exits

For reference, the nodes are labelled left to right:

Node	Role	Generates traffic	Links to
S1	Edge node (top left)	Yes	A
S2	Edge node (middle left)	Yes	A
S3	Edge node (bottom left)	Yes	B
A	Inner node	No	S1, S2, B, C
B	Inner node	Yes	S3, A, C, D
C	Inner node (upper)	Yes	A, B, UK, US
D	Inner node (lower)	No	B, UK, US
UK	Exit	No	C, D
US	Exit	No	C, D

Every link is full-duplex and has its own bandwidth. The sketch gives one example: the S1–A link carries 30 Mbps with 20 ms latency. The dashed route shows one possible routing choice: traffic from S1 that reaches US by going down through B instead of across through A and C.

The network in the source PDF is a simpler variant (8 nodes, one hub H in front of both exits). The problem statement covers both; nothing in it depends on the layout.

Traffic

At any instant the network carries exactly four streams. Each has a fixed exit and a current demand, its rate at that instant summed over the source nodes it comes from.

Stream	Exit	Latency-sensitive	Allowed paths
UK-L	UK	Yes	Any path to UK; latency should be as low as possible
UK-T	UK	No	Any path to UK
US-L	US	Yes	Any path to US; latency should be as low as possible
US-T	US	No	Any path to US

A path's latency is the sum of the latencies of its links. For UK-L and US-L, lower latency is better, with no fixed cut-off. For UK-T and US-T, latency does not matter.

Constraints

Any routing of the current demand must satisfy all of the following:

1. **Link bandwidth.** The four streams together stay within each link's bandwidth, in each direction.
2. **Exit.** Each stream leaves only at its own exit: UK-L and UK-T at UK, US-L and US-T at US.
3. **Demand.** No stream is routed more than its current demand. Demand that cannot be routed is lost.
4. **Path rule (variant).** Either each stream stays on one path, or its traffic may be split across several. Both variants should be solved.

Objective

The routing has two goals at the instant being solved:

- **Throughput:** deliver as much of the four streams' total demand as possible.
- **Latency:** keep the latency of UK-L and US-L as low as possible.

There are no fixed latency limits. The fastest paths have limited bandwidth, and the goals can pull against each other: sending latency-sensitive traffic only on the fastest paths may leave demand undelivered, and filling them with tolerant traffic raises latency for the sensitive streams. How to define and balance the two goals is part of the problem for the candidate to work out. When demand changes, the problem is solved again for the new instant.

Assumptions

ID	Assumption
A1	Each problem is a snapshot: demand is fixed at its current value, with no change over time.
A2	There are exactly four streams: one latency-sensitive and one tolerant stream per exit.
A3	A path's latency is the sum of its link latencies. There is no queuing or waiting at nodes.
A4	Links are full-duplex and lossless. Only link bandwidth limits how much traffic a path carries.
A5	All delivered traffic is weighted equally.
A6	Traffic leaves the network only at UK and US.

All bandwidths, latencies and demands are illustrative until real figures are available.

Inputs and expected outputs

Inputs

- The graph: nodes, links, and the two exits (UK and US).
- For each link: bandwidth per direction and latency.

- For each of the four streams: its source nodes, its current demand.

Outputs

- For each stream: the demand delivered and the path, or split across paths, it uses.
- For each stream: the demand that could not be delivered.
- The total delivered across all four streams.
- The bottleneck links, and how much one more unit of bandwidth on each would add.